

EPSRC Research Proposal

Business Understanding

- Shadow doctor appointments within the LTHT
- Review the current state of the EHR system

Data Understanding

Analyse pre-existing clinical documentation

- Obtain access to the NHS's SDE
- Review available data on the NHS's SDE
- Create a standardised clinical summary document
- Collect samples of patient-doctor conversations from LTHT
- Completion of data collection phase from LTHT

Data Preparation

Development of data pre-processing program

- Implementing locally runnable robust clinical STT
- Design and develop patient pseudonymisation program
- Development of STT preprocessing pipeline
- Process patient-doctor conversations through program

Modelling

GPT related

- Gain access and learn about the GPT-4 API
- develop standardised clinical summary document
- Process pseudonymised conversation
- Integration of GPT-4 Turbo API for summarisation
- Develop action mapping algorithm
- Implementation of action mapping algorithm for treatment plans

Evaluation

- Development of software package as a runnable application
- Stakeholder feedback through user testing
- Refinements to application
- Deliverable software prototype

Proposal for Integration

- Investigate methods of integration into NHS workflow
- investigate additional training required for use
- investigate additional essential areas the software needs to cover
- investigate implementation of action mapping to NHS systems

Report development

- Develop report
- Completion of final report
- Create presentation
- Completion of presentation
- Create publication for peer-reviewed journal
- Completion of publication

